

Reflection Paper: Data Analytics Internship at New Dia Cannabis Co. – Fall 2025 Update

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Since my original BA890 summer reflection, my role at New Dia has evolved from “intern building dashboards” into something closer to an in-house operational analytics, consumer insights and pricing strategist. From September through November, I stayed on with New Dia while returning to MSBA classes, and my scope shifted in three big ways:

1. owning the core data and KPI infrastructure,
2. leading deeper program analytics for discounts and loyalty, and
3. translating those insights into concrete forecasts, goals and experiments for the Fenway and Worcester stores.

This paper is an update that focuses on those new responsibilities and what I learned from them.

1. Owning the KPI and data infrastructure

The first major change was that I became the person everyone relied on to make the numbers “truth.” What started as simple weekly reports turned into a full redesign of New Dia’s operational KPI stack.

I took over weekly performance analytics reports twice a week: a Monday lookback and a Thursday “rest-of-week” outlook. These reports covered net sales, transactions, ATV, units per transaction, category mix and discount behavior, and they became the backbone for our store huddles.

To make those reports trustworthy, I had to fix the underlying data. That meant:

- **Cleaning core dimensions:** I found that “New customer KPIs” were inaccurate due to classification logic in the transaction feeds, so I rebuilt the new vs returning flags and refreshed all downstream dashboards. Similarly, “day of week” fields were mis-assigned, so I corrected them and enforced a consistent week structure that starts on Monday and ends on Sunday, which aligns better with how the team thinks about staffing and promotions.
- **Aligning conflicting dashboards:** There were data mismatches between the internal New Dia sales dashboard and the CASA sales dashboard. I traced those back to

missing filters and inconsistent time windows, then reconciled the logic so that both tools now agree on net sales, transactions and margin. I also standardized weekday KPI tables so they align across reports.

- **Enriching the Dutchie transaction table:** I worked with our AWS Casa QuickSight environment to add more “intelligence” to each row of the transaction table. Concretely, I:
 - Included **discount name** on each transaction line and mapped it to a higher level **discount group** so we can analyze Add-On, Marketing, Loyalty and other program families consistently.
 - Added **budtender ID** so we can analyze performance at the staff level.
 - Built **customer transaction groups** and flags for “discounts except” logic so we can isolate impact of specific programs.
 - Fixed **pricing tier mismatches** and added an **infused vs non-infused** filter, which is critical for pre-roll and vape analysis.
 - Added **unit count** and **transaction count filters**, plus **customer count** to weekly and monthly KPI tables, so we can move smoothly between revenue and people-based views.
 - Included **hours** in weekly and monthly pivot tables, which later supported hour-by-hour performance, wait-time vs sales analysis and staffing decisions.
- **Building more useful filters and tools:** I created a dedicated sheet of **cart-level discounts** and a **net sales range filter**, which allowed managers to quickly isolate high-value baskets or specific promo tiers. I also built a **wait time vs sales filter** to start exploring queue-length elasticity.

This work moved me from “someone who builds charts” into “owner of the source of truth.” It taught me that analytics power really starts once the business trusts the base table, and it also forced me to practice data governance, not just SQL.

2. Deep-diving into Pledge Dia and discount economics

The second major shift was taking full ownership of analytics for Pledge Dia and other discount programs, especially in the student segment. This work sat at the intersection of basket analysis, elasticity, CLV and acquisition economics.

Basket and attach analysis for Pledge Dia

Using a transaction table filtered to Pledge Dia transactions, and customer segments from our Alpine IQ loyalty platform, I performed a full **basket and attach analysis** on student-tagged customers. I:

- Built a **product-level attach matrix** to understand what students buy on top of the \$1 pre-roll.
- Calculated **attach rates by category** and identified which categories (for example apparel, accessories, infused pre-rolls and vapes) drive profitable cross-sell beyond the initial discount.
- Ranked complementary SKUs by **incremental margin per student**, focusing on combinations where lift was above 1.2 and margin remained positive.

Because it costs us approximately **\$1.60 to get a student to the door**, I framed every Pledge Dia metric through an acquisition lens: Are we beating the \$1.60 CAC, and over how many visits?

RFM and rule-based personas for students

To go beyond one-time baskets, I computed **RFM scores for student customers** and used rule-based personas to surface “daily micro-buyers” versus “weekend stock-ups.” I:

- Calculated Recency, Frequency and Monetary values per student and mapped them into strategy cells (for example, 555 loyalists versus 155 churn-risk samplers).
- Separated **one-time buyers from repeaters** and then measured CAC payback by dividing the \$1.60 acquisition cost by gross profit per visit.
- Evaluated **acquisition efficiency** in terms of cost per retained student, not just cost per visit.

This helped quantify which student behaviors justify deeper discounts and where we should instead protect margin.

Comparative discount behavior within Pledge Dia

Within the broader Pledge Dia universe I also:

- Compared **customers who redeemed “New Dia Pledge 10 percent off”** versus those who primarily used other discounts (sometimes stacked on top of Pledge). I examined

basket size, category mix and redemption frequency per month for both segments.

- Compared **customers who buy a single \$1 pre-roll** versus those who buy **ten pre-rolls for \$30**, looking at units per transaction, add-on behavior and how they evolve over time.
- Looked specifically at **Pledge Fathom 3 dollar joints** performance and how that price point shapes units, revenue and repeat behavior.

This work translated classroom topics like elasticity of demand and segmentation into very concrete questions: Which price points and structures maximize both units and lifetime margin, not just one good promo day.

Outage, lost revenue and recoup modeling

When we had a **3.5-day outage**, I carried out a structured **lost revenue analysis**:

- Built daily index charts for **Net Sales, Transactions and ATV** for the four weeks surrounding the outage and flagged the closure window plus the first three days post-reopen.
- Calculated how much of the lost volume was recaptured on Wednesday through Friday, expressed as a **gap recoup percentage**.
- Built a simple **time-to-recoup model** to estimate how many days it would take to recover the lost revenue given the uplift observed after reopening.

I then connected that to **price and discount effects**, looking at discount redemption rate pre-versus post-reopen and running a **margin impact simulation** where I re-priced baskets at regular price to estimate gross margin dilution from make-good discounts. This helped answer why Wednesday and Thursday overperformed relative to typical weeks and what those customers were buying after reopening.

3. MVP program, CLV and forecasting work

In parallel, I started working more with our loyalty and MVP programs, bridging RFM, clustering and CLV ideas from BA881 into real store decisions.

MVP discount and RFM + K-means segmentation

For the MVP membership base, I:

- Ran **reward schedule analysis** to see how MVP discounts actually ladder across tiers and whether they match realized value.
- Built **Recency-Frequency-Monetary metrics** for every MVP member who transacted between 18 August and 27 October 2025.
- Standardized these metrics and ran **K-means clustering with k = 4**, which produced intuitive tiers:
 - Platinum (about 31 very high-value VIPs)
 - Gold (about 430 loyalists)
 - Silver (about 5k steady buyers)
 - Bronze (about 4.3k dormant or low-value members)

This gave us a data-driven way to align discounts and experiences with true economic value, instead of treating “MVP” as a single homogeneous group.

Fenway event-based goal setting and daily targets

For the Fenway store, I helped build **September October and November sales projections and daily goals** using previous month goals as a reference and layering in Red Sox home games and concert events.

Key design choices included:

- Setting a **net month target of approximately 800K dollars** and then backing into day-level goals.
- Locking the **Avg Txn Goal at 48 dollars**, about 15 percent above the September actual of approximately 41.40, to incentivize upsell discipline without assuming unrealistic discounting.
- Using Fenway patterns to estimate **Est. number of transactions**, differentiating Monday to Thursday base days, weekend lift, and game or concert surges. We then scaled these patterns so that the month-end sum landed at about 800K.
- Deriving **ID scan goals** by assuming an 80 percent conversion from scan to transaction, so $\text{Id Scan Goal} \approx \text{Est. Transactions} \div 0.8$. This led to daily ID scan goals of roughly 1,000 per day.

This exercise forced me to integrate **pricing, traffic and operational constraints**. For example, pushing to 1,000 transactions per day at 48 dollars average spend is not realistically achievable without heavier discounting or a major traffic shock. We deliberately chose a more balanced combination of volume and basket growth instead of chasing vanity transaction counts. It also highlighted non-technical levers like budtender training, “fast-lane” signage for pre-scanned customers, and scanning every guest in multi-ID groups.

Lifetime value and mail-to-home flyer discounts

Another new frontier was thinking about **lifetime value for customers acquired via mail-to-home flyers** featuring offers such as:

- “Woo – 40 percent entire order”
- “Woo – 50 percent off one item”
- “Woo – 25 dollars off 60 dollars or more”

Using the last 180 days as a frame, I outlined an approach to drop this data into a **BG/NBD or Pareto/NBD style model** to project full CLV curves for these cohorts. The core questions were: What loop and conversion rates do we see today, and what would it take to convert these one-time deal seekers into loyalists?

I also:

- Tagged each **first visit with a discount group** (Add-On, Loyalty, Marketing, etc.) to see which archetypes over- or under-convert to second visits.
- Appended **marketing source identifiers** (for example, Surfside vs walk-by) to quantify CAC-to-repeat ROI.
- Pulled **Alpine IQ loyalty status** and sign-up timing into this view to see whether earlier program enrollment shortens the lag to second visit.

This work laid the foundation for a more rigorous CLV framework at New Dia, grounded in channels and discount families, not just aggregate averages.

4. Broader portfolio of analyses and customer insights

Beyond the flagship discount programs, I also delivered a broad set of targeted analyses that deepened my understanding of cannabis retail behavior:

- Built weekly **WOW apparel and accessories line charts**, treating these higher-margin items as attach anchors and watching for trend inflections.
- Owned the KPI and Analytics Layer for **Delivery** performance and customer behaviour.
- Analysed the **Event Driven Spikes** to understand the flow of customers from external events in Fenway and built a prediction model to analyse decision behaviour.
- Explored **correlations between 1oz Fresh Pack performance and other weeks**, to understand whether heavy 1oz promotions cannibalize future demand or simply pull forward volume.
- Analyzed the **secret menu** and how it attracts specific segments and price tiers.
- Examined **21–24 year old customers at the Worcester store**, using addresses, products and behavior to understand how young adults shop differently and which discounts or product types resonate with them.
- Conducted **pricing tier analysis** across categories to see where laddering is coherent versus confusing or misaligned with demand.
- Looked at **repeat customers at the gas station location**, focusing on who visits most frequently, how many visits they make, and which categories (including 2 g vapes) they favor.
- Quantified **how many pre-rolls customers buy: singles vs multipacks**, linking it to units per transaction and promotion structure.
- Built a **zipcode heatmap** to visualize geographic demand concentration and support customer journey mapping from neighborhood to store.
- Worked with the Associate Operations Director to Define Inhouse Budtender monetary and product level Incentives to boost healthy competition and increase sales based on customer purchase patterns per weekday.
- Analysed In house Brands - **Gate 420, Smoke Show, Hall of Flame KPIs** regularly
- Produced a vendor performance report for **Sluggers**, evaluating velocity, margin and repeat behavior by product.
- Analyzed **Terpli reports on ad performance**, exploring touchpoint attribution across channels and how ad exposure links to in-store spending patterns.

Together, these projects trained me to think not as a student running isolated analysis, but as a retail strategist asking “If this is true, how does it change what we put on the menu, how we price it and how we allocate discounts and budget?”

5. Learning and impact

From a learning perspective, this fall felt like an applied extension of my MSBA coursework. Concepts from elasticity, segmentation, CLV, customer choice, and discount optimization stopped being abstract and became daily decision tools:

- Elasticity appeared in **price mocks** where I simulated plus or minus 1 dollar changes on 3-packs and 5-packs to see revenue versus margin tradeoffs before holiday repricing.
- Segmentation went from theory to practice through **RFM cells, student personas and MVP K-means clusters** that directly informed discount depth and messaging.
- CLV shifted my focus from daily sales to **CAC payback and repeat profitability**, especially in programs where it costs us 1.60 dollars to get a student in the door.
- Operational analytics turned into a discipline of making sure **every dashboard reconciles**, every field is defined once, and every KPI table aligns with how managers actually run the store week by week.

Most importantly, I experienced how analytics is not about “being clever with data,” but about creating systems that the business can run every week: cleaned tables, consistent dimensions, clear goals and recurring reports that connect to real decisions.

By November, I felt I had moved from simply describing what happened, to helping explain why it happened, and, increasingly, to suggesting what New Dia should try next. That progression, from reporting to diagnosis to prescription, is exactly the kind of trajectory I hoped for when I joined the MSBA program, and it has shaped how I now think about my career as a consumer insights and analytics leader.